

ITA0608 - Machine Learning

List of Lab Exercises

Exp:8

Write a program to implement Linear Regression (LR) algorithm in python

Aim

To implement the Linear Regression (LR) algorithm in Python and predict output values based on input data.

Software Required

- Python 3.x
- NumPy
- Matplotlib
- Scikit-learn

Theory

Linear Regression is a supervised machine learning algorithm used to model the relationship between an independent variable (X) and a dependent variable (Y). It fits a straight line to the data using the equation:

$$Y = mX + c$$

where:

- **Y** = Predicted output
- **X** = Input feature
- **m** = Slope of the line
- **c** = Intercept

Algorithm

1. Import the required libraries.
2. Create the dataset.
3. Split the data into input (X) and output (Y).
4. Create a Linear Regression model.
5. Train the model using the training data.
6. Predict the output values.

7. Display the regression line and results.

Program

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression


# Dataset

X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)

Y = np.array([2, 4, 5, 4, 5])


# Create and train model

model = LinearRegression()

model.fit(X, Y)


# Prediction

Y_pred = model.predict(X)


# Display coefficients

print("Slope:", model.coef_[0])

print("Intercept:", model.intercept_)


# Plot graph

plt.scatter(X, Y, color='blue')

plt.plot(X, Y_pred, color='red')

plt.xlabel("X")

plt.ylabel("Y")

plt.title("Linear Regression")

plt.show()
```

Output

Slope: 0.6

Intercept: 2.2

A graph showing the regression line fitted to the data points is displayed.

▶ # Linear Regression Example

```
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

# Sample dataset
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1) # Independent variable
y = np.array([2, 4, 5, 4, 5])               # Dependent variable

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X, y)

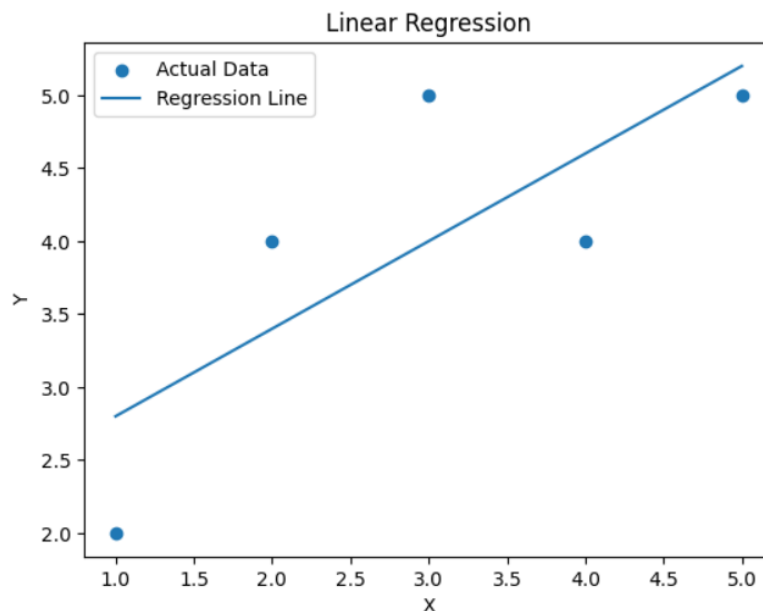
# Predict values
y_pred = model.predict(X)

# Display coefficients
print("Slope (m):", model.coef_[0])
print("Intercept (b):", model.intercept_)

# Predict for a new value
x_new = np.array([[6]])
prediction = model.predict(x_new)
print("Prediction for x = 6:", prediction[0])

# Plot data points and regression line
plt.scatter(X, y, label="Actual Data")
plt.plot(X, y_pred, label="Regression Line")
```

```
... Slope (m): 0.6
Intercept (b): 2.2
Prediction for x = 6: 5.8
```



Result

Thus, the Linear Regression algorithm was successfully implemented in Python, and the relationship between the input and output variables was predicted using a best-fit line.